



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI THAYARS AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

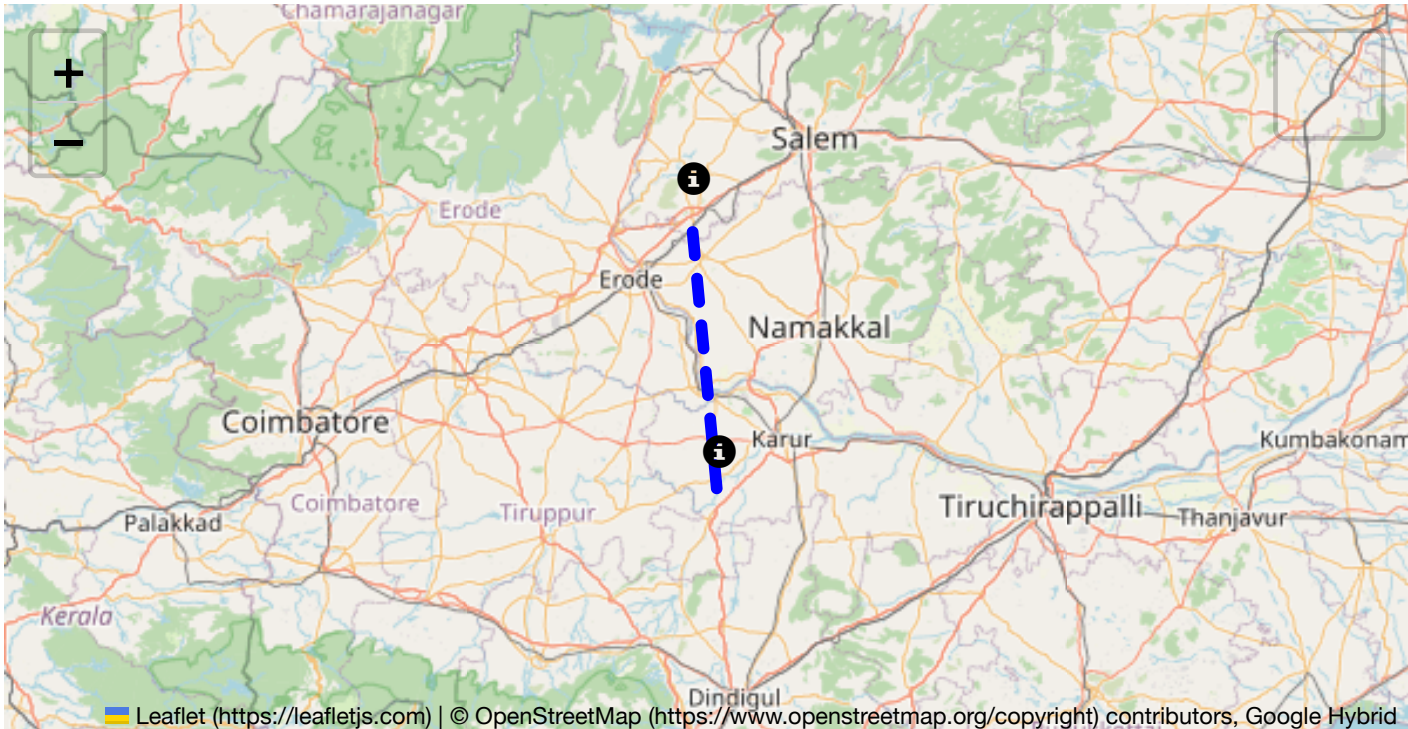
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

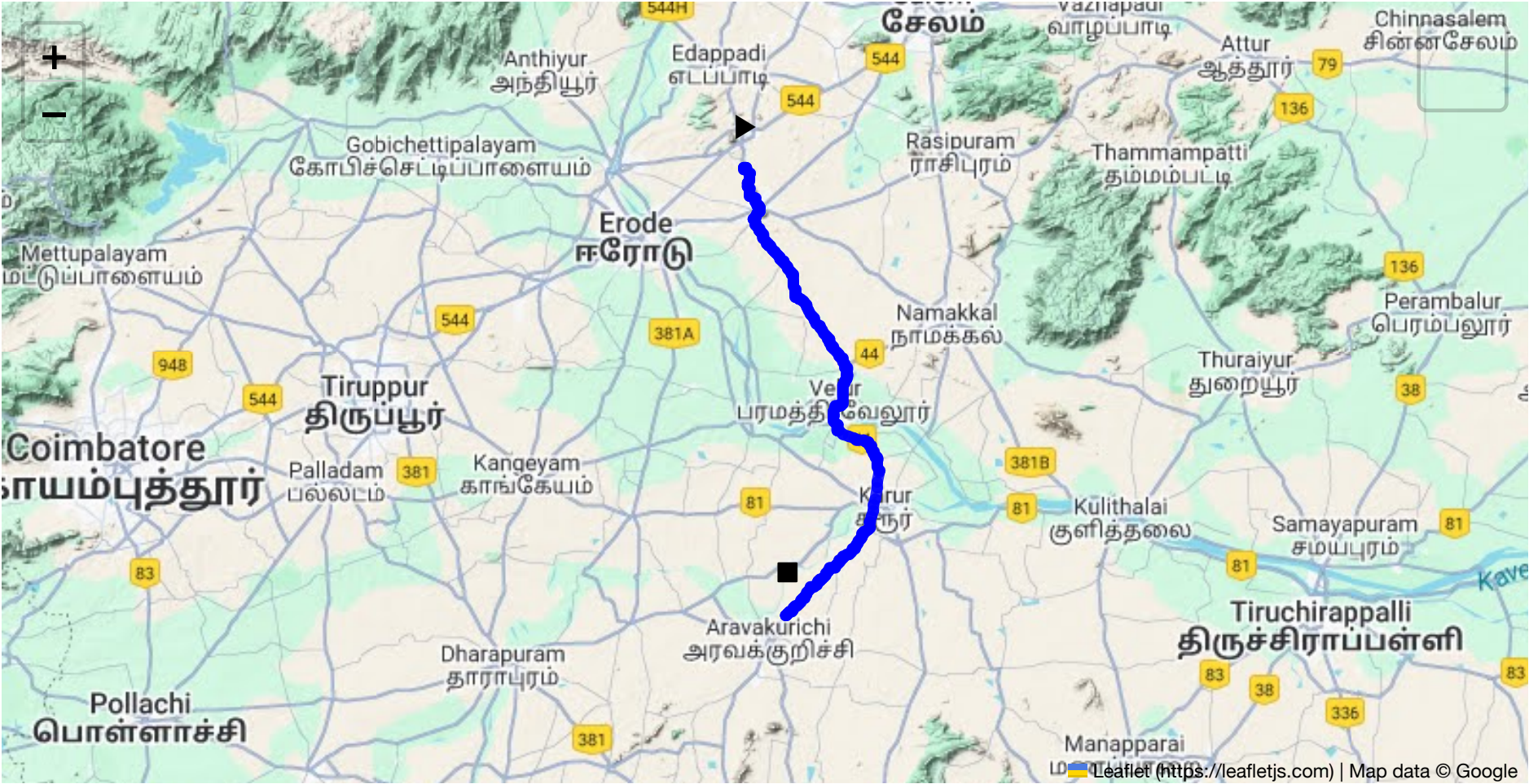
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 87.60 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours

Start: (11.4381, 77.8734)
End: (10.806362, 77.934242)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Pungambadi West

1. Overview of the Route Map

The route from Sangagiri (CVQF+23W) to Pungambadi West (RW4M+VJ8) is approximately 87.60 kilometers via National Highway 44 (NH44) and regional roads. This route typically connects key rural and semi-urban regions of Tamil Nadu, offering a mix of highways and local roads.

2. Typical Weather Conditions and Potential Hazards

Tamil Nadu experiences a tropical climate with considerable seasonal variation:

- **Monsoon Season (June to September):** Heavy rainfall can lead to waterlogged roads and landslides in elevated areas.
- **Summer (March to June):** High temperatures can cause vehicle overheating and affect asphalt road surfaces.
- **Winter (December to February):** Though milder, occasional fog can reduce visibility, especially in the early mornings.

3. Traffic Patterns and Congestion Areas

- **Peak Hours:** Morning (7 AM - 10 AM) and evening (5 PM - 8 PM) see increased traffic due to daily commuting.
- **Congestion-Prone Areas:** Entry and exit points of NH44 can experience bottlenecks, particularly near markets and major intersections in towns like Namakkal.

4. Road Quality and Infrastructure

- **NH44** is generally well-maintained, with clear marking and signage.
- **Secondary roads** may vary in quality, often narrower with potential potholes and limited shoulder space.
- Bridges and overpasses are structurally sound but should be approached with caution during heavy rain due to slipping hazards.

5. Alternative Routes for Emergencies

- **Alternative Route:** In case of a blockage on NH44, SR132 offers a parallel path but may add time due to its two-lane configuration.
- Stay updated on local traffic apps or radio announcements for real-time diversions.

6. Local Regulations Affecting Hazardous Material Transport

- **Curfews:** Some towns may enforce strict timings for heavy vehicle entry to avoid congestion.
- **Permits:** Ensure compliance with Tamil Nadu's regulations for hazardous materials, requiring proper documentation and safety measures.

7. Historical Incidents

- **Accident Hotspots:** Certain intersections near Namakkal have reported higher accident rates, typically involving vehicle collisions.
- **Material Spillage:** Rare incidents of hazardous material spillage emphasize strict adherence to safety protocols.

8. Environmental Considerations and Sensitive Areas

- Avoid using roads in proximity to agricultural lands and water bodies to prevent contamination in case of spillage.
- Areas near the Cauvery River and reserved forests should be traversed with due environmental caution.

9. Communication Coverage

- **Network Stability:** Adequate cellular network in urban zones, but potential dead zones exist in rural stretches, especially near forested areas.
- Consider using GPS devices with offline capabilities.

10. Estimated Emergency Response Times

- **Urban Areas (e.g., Namakkal):** Approximately 20-30 minutes due to available local emergency services.
- **Rural and Remote Areas:** Response times can extend up to 60 minutes based on road conditions and availability of nearest emergency units.

11. Overall Risk Assessment Summary

The route presents a moderate risk level for transporting hazardous materials due to a blend of high-quality infrastructure with seasonal weather-related challenges. Regular maintenance and adherence to

safety protocols are crucial. Awareness of local traffic patterns and environmental considerations will further mitigate potential hazards.

Conclusion

Drivers should receive thorough training in handling adverse weather conditions, emergency response preparation, and regulations. Ensuring vehicle readiness and real-time communication can enhance safety during transit on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr
8	Turn	Medium	11.15579, 78.02169	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14485, 78.02282	15 KM/Hr
11	Turn	High	10.80699, 77.93429	15 KM/Hr
12	Turn	High	10.80676, 77.93437	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
21	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15579, 78.02169



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14485, 78.02282



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.80699, 77.93429



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.80676, 77.93437

